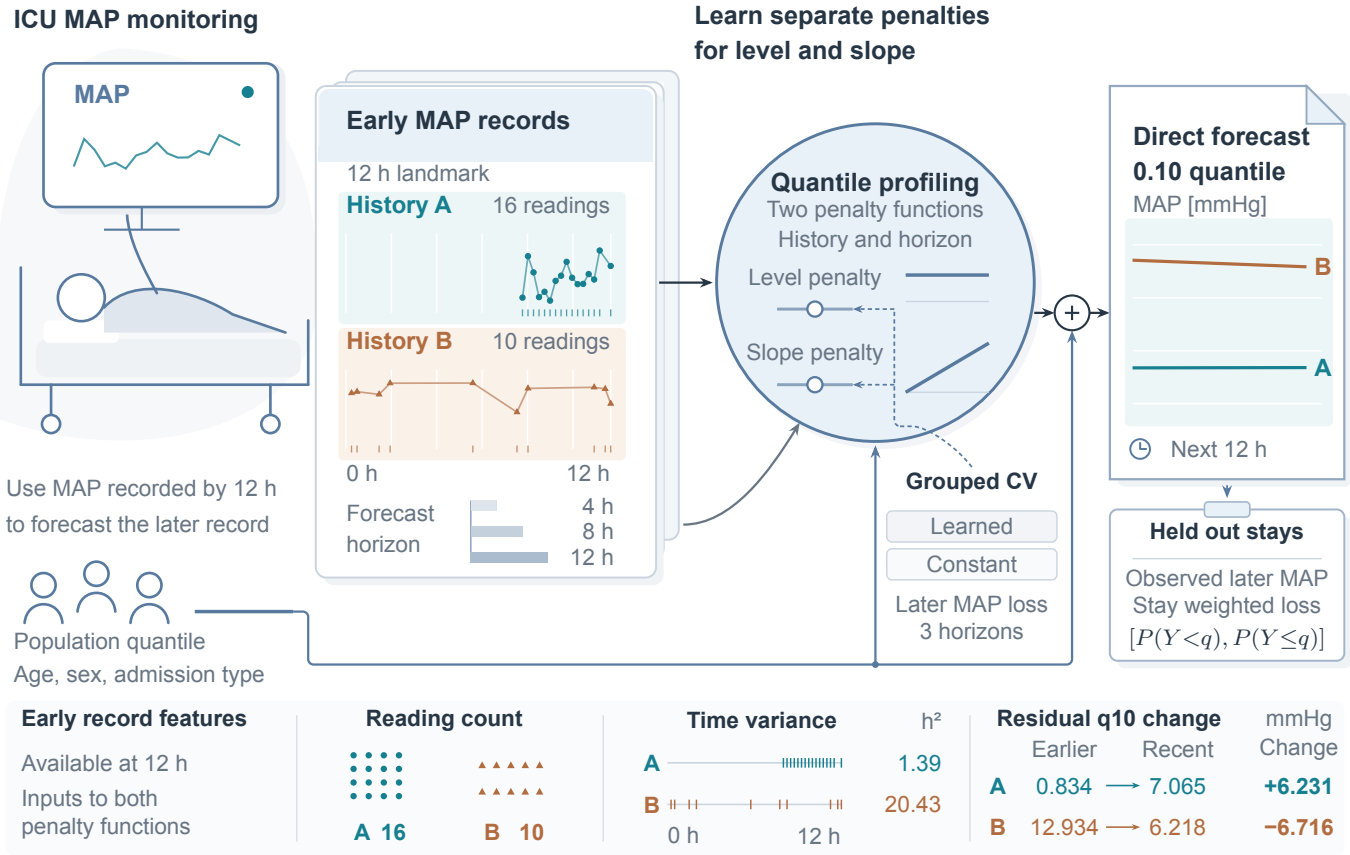

ICU MAP monitoring
MAP
Use MAP recorded by 12 h
to forecast the later record
Population quantile
Age, sex, admission type
Early MAP records
12 h landmark
History A
16 readings
History B
10 readings
0 h
12 h
Forecast
horizon
4 h
8 h
12 h
Learn separate penalties
for level and slope
Quantile profiling
Two penalty functions
History and horizon
Level penalty
Slope penalty
Grouped CV
Learned
Constant
Later MAP loss
3 horizons
Direct forecast
0.10 quantile
MAP [mmHg]
B
A
Next 12 h
Held out stays
Observed later MAP
Stay weighted loss
$[P(Y<q), P(Y\leq q)]$
Early record features
Available at 12 h
Inputs to both
penalty functions
Reading count
A 16
B 10
Time variance
h²
A
1.39
B
20.43
0 h
12 h
Residual q10 change
Earlier
Recent
mmHg
Change
A 0.834 → 7.065 +6.231
B 12.934 → 6.218 −6.716
Shared learning
$\mathcal{F}$: population quantile
$\mathcal{T}$: 3-fold CV selects learned or constant penalties